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The role of spectral characteristics of urine in bladder cancer diagnostics

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Finding a non-invasive diagnostic method with sufficient diagnostic power is crucial for early detection of malignant tumor diseases. The main goal of the presented work is to observe changes in the spectral characteristics of urine between patients diagnosed with bladder cancer and control subjects. Data were obtained through fluorescence spectroscopy and high-performance liquid chromatography (HPLC). The data obtained from multiple fluorescence spectra measurements were graphically represented as excitation-emission matrices (EEMs). In both EEMs and chromatograms, statistically significant peaks and areas were identified, which were evaluated using various statistical methods and machine learning techniques (logistic regression, OPLS-DA, convolutional neural networks). The analysis of urine EEMs did not yield satisfactory results; the highest accuracy was achieved using convolutional neural networks, with a maximum accuracy of 72.1% for the training model. Regarding chromatograms, the best results were obtained by applying convolutional neural networks to chromatogram data, achieving an accuracy of 95.3% for the training model. Established methods of data standardization did not improve performance of discrimination models. Our findings highlight the importance of a multi-parametric approach that captures interactions among spectral features, reflecting not only the complexity of cancer but also inter-individual variability among patients. The integration of urinary spectral data with advanced machine learning methods shows potential for improving patient stratification in BC.

Keywords Urine, Fluorescence, Absorption, Chromatography, Machine learning, Bladder cancer

According to the latest available data from the International Agency for Research on Cancer (IARC), bladder cancer (BC) caused the death of more than 220,000 people worldwide in 2022[1]. The IARC also states that this type of cancer is very financially demanding in terms of diagnosis and treatment, mainly because of its high recurrence and progression rates. It is known that the sooner the cancer is diagnosed, the better the prognosis there is for the patient[2]. Nowadays, diagnostics mainly consist of patient´s anamnesis, followed by various physical and laboratory examinations such as invasive cystoscopy and urine cytology, both of which have limitations. Cystoscopy is an invasive procedure and urinary cytology comes with heterogeneity in interpretation among uro-pathologists, difficulties in interpretation in case of urinary tract infection, and low sensitivity especially for low-grade tumors[3,4].

Urine-based non-invasive screening tests have become a hotspot of current research in recent years. Since urine is in direct contact with the affected tissue, it reflects the current state of the bladder[5]. Several commercial molecular tests have been approved by the Food and Drug Administration (FDA) for surveillance to avoid repetitive cystoscopy. These tests include the measurement of soluble proteins such as bladder tumor antigen (BTA Stat, BTA TRAK), nuclear matrix protein 22 (NMP22 BC, NMP22 BladderChek), proteins detected on urothelial cells (ImmunoCyt™), or chromosomal aberrations (UroVysion®). However, their limited sensitivity and specificity makes it difficult to use them effectively in clinical practice[6]. Of all the biomarkers studied so far, only four have achieved specificity and sensitivity values above 90%. Multi-target biomarkers perform slightly better, but to this day, no biomarkers are known that could replace cystoscopy[7].

It is known that cancer cells alter the metabolism, leading to modified composition of body fluids (including urine) in patients with BC. Different metabolites have different spectral characteristics, which can be used to

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compare urine spectral profiles between patients with and without BC. Based on this assumption, changes in the metabolite profile could form the basis for non-invasive untargeted metabolomics.

Experimental methods used for investigating complex body fluid samples produce big data, mainly in terms of multi-omic experiments (genomic, transcriptomic, proteomic, metabolomic, etc.) but also from spectral and chromatographic measurements. The volume and complexity of relationships that big data reflect require new methods and technologies for their processing[8]. In case of spectroscopy, it is appropriate to use multivariate methods or machine learning (ML) for evaluating the results. Both artificial intelligence subsets represented by machine learning (ML) and deep learning (DL) techniques are heavily studied regarding the diagnosis, prognosis, and prediction of outcomes of BC[9]. In this work, we aim to apply multivariate statistical analysis and ML techniques to the spectral properties of urine obtained from two groups of subjects using fluorescence spectroscopy and HPLC. Spectroscopy combined with ML has the potential to detect early stages of cancer compared to current diagnostic methods, and could thus replace invasive cystoscopy in the future[9].

Results

Excitation-emission matrices

The output from the fluorescence measurement is represented by three-dimensional excitation-emission matrices (EEM, Fig. 1), which show both the excitation and emission spectra in the wavelength range of 200–800 nm. Measured data were processed and evaluated using statistical methods used in untargeted metabolomics, the main goal of which is to study the metabolic composition of cells, tissues, or biological fluids.

Fluorescence peak analysis

Excitation-emission matrices obtained by measuring the excitation-emission spectra of urine entered the peak analysis. Due to missing information on creatinine, 21 samples were excluded. The missing data were of the MCAR type (missing completely at random;[10]. Data were subjected to univariate analysis, ROC analysis, logistic regression, and OPLS-DA.

Univariate analysis Using multiple measurements of fluorescence spectra and univariate analysis, nine peaks were identified, none of which were significantly different between patients with BC and controls (Supplementary Table S1; $p < 0.05$). On the contrary, after standardization to creatinine all peaks were statistically significant ($p < 0.05$).

ROC analysis For both unstandardized and standardized peaks, the accuracy of the ROC analysis for individual peaks was in the range of 52–58% and 58–63%, with sensitivity from 36 to 94% and 44–82%, and specificity from 14 to 76% and 38–77%, respectively. The area under the curve (AUC) reached a minimum value of 0.5 and

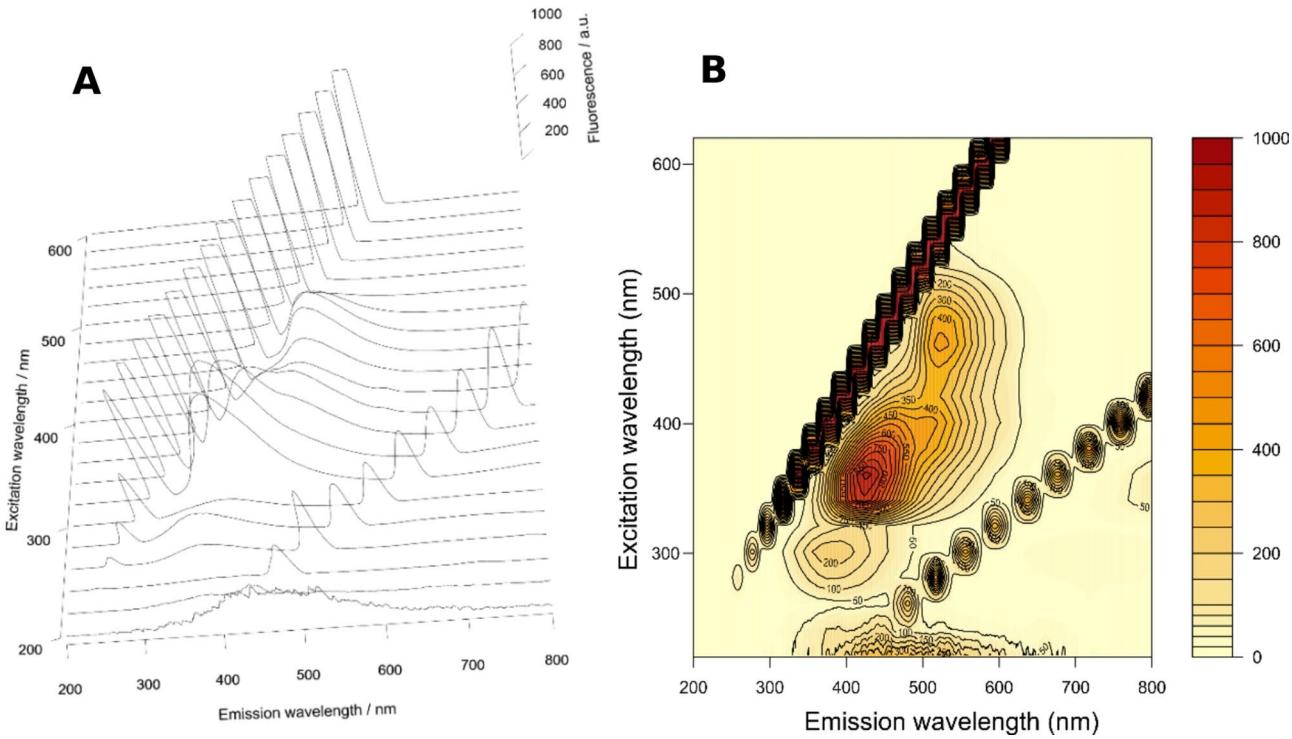

Fig. 1. Representative fluorescence spectra of urine as (**A**) 3d lines plot and (**B**) contour plot (EEM). The contour plot shows the signal using a color scale.

0.6 and a maximum of 0.54 and 0.65. The results of four selected peaks with the best AUC values are summarized in Supplementary Table S2.

Logistic regression All identified peaks entered the logistic regression model. For unstandardized peaks and peaks standardized to creatinine the most important was peak 4 (excitation/emission 440/520 nm). This peak was selected based on the lowest BIC (Bayesian Information Criterion). Sensitivity, specificity, accuracy, and AUC for unstandardized peaks were 52.6% (43–63%), 52.8% (42–63%), 52.7% (46–60%) and 0.54 (0.46–0.62), respectively. For standardized peaks, sensitivity, specificity, accuracy, and AUC (95% CI) were 73.6% (63–82%), 44.1% (34–55%), 58.3% (51–66%) and 0.63 (0.55–0.71), respectively.

In addition to these parameters, we included the parameters "gender" and "age" in the model since none of the peaks reached a VIP value > 0.8. After adding the parameters "age" and "sex" to the model, for unstandardized peaks these two parameters showed the greatest weight (no peak was included in the model). When using standardized peaks, the best model was achieved by combining parameters "gender" and "age" with peak 4. Sensitivity, specificity, accuracy, and AUC (95% CI) for unstandardized peaks were 62.1% (53–73%), 67.9% (59–78%), 65.2% (59–73%) and 0.68 (0.61–0.76), respectively. For peaks standardized to creatinine, sensitivity, specificity, accuracy, and AUC were 68.9% (58–78%), 62.4% (52–72%), 65.6% (58–72%) and 0.71 (0.64–0.79), respectively.

OPLS-DA The OPLS-DA method was used to distinguish diagnostic groups, and in the next step the sensitivity, specificity, and accuracy of the created model were calculated. The OPLS-DA separates predictive variation related to class discrimination (explanatory component) from non-predictive variation (orthogonal, within-group variance) in datasets of variables (peaks intensity). The results of OPLS-DA analysis for both types of peaks are presented in Table 1. Validation of the model was performed using seven-fold cross-validation.

Fluorescent area analysis

Univariate analysis The obtained EEM were divided into 372 areas, from which areas 253–372 were excluded due to no informative value (excitation wavelength above 600 nm), as well as areas of Rayleigh scatter (51 areas). In total, 201 unstandardized samples, but only 180 standardized ones (missing information on creatinine) entered the univariate analysis and OPLS-DA.

Ten statistically significant areas ($p < 0.05$) that differed between controls and BC patients were identified. After standardization to creatinine, up to 92 areas were statistically significant (Supplementary Fig. S1). Since the areas in Rayleigh scattering were also statistically significant, we omitted them from further analyzes.

ROC analysis For unstandardized areas, the accuracy of ROC analysis for individual areas was in the range of 44–67%, with sensitivity from 4 to 100% and specificity from 4 to 100%. The area under the curve (AUC) reached a minimum value of 0.50 and a maximum of 0.60. ROC analysis for standardized areas reached an accuracy of 56–76%, sensitivity of 34–96% and specificity of 23–89% with AUC values ranging from a minimum of 0.50 to a maximum of 0.74. Five areas with the highest AUC values were selected (Supplementary Tab. S3).

OPLS-DA Unstandardized as well as standardized areas were subjected to OPLS-DA analysis in two ways: (1) all fluorescent areas included; (2) non-colinear areas less than 0.9 included.

These two types of analysis were compared. Two components (one explanatory and one orthogonal) entered the final models. In case areas with excluded mutually correlated parameters ($r > 0.9$), two orthogonal components were used. In Table 2, the results of OPLS-DA are presented and compared.

Convolutional neural networks (CNN) Using CNN, two models based on urine fluorescence changes that could distinguish BC patients from controls were created. In the first case, the model was trained on all samples (training model). In the second case, a 10-fold cross-validation took place (validated model). The number of epochs in both models was 30. The settings of both models are listed in Supplementary Table S4. The training model achieved an accuracy of 72.1%, with a sensitivity of 90% (95% CI 83–95%) and a specificity of 51% (95% CI 40–62%). With the validated model, accuracy dropped to 70%, specificity to 30% (95% CI 20–41%), but sensitivity increased to 92% (95% CI 86–97%).

Std.	R2X	R2Y	Q2	pR2Y	pQ2	Sensitivity (95% CI)	Specificity (95% CI)	Accuracy (95% CI)
NP	56.7%	13%	0.07	0.05	0.05	64.2% (53.7–73.8)	66.9% (57.2–75.8)	65.7% (58.7–72.2)
Cr	80.6%	13.1%	0.09	0.05	0.05	67.8% (56.9–77.4)	68.8% (58.4–78)	68.3% (60.9–75.1)

Table 1. OPLS-DA results for unstandardized peaks (NP) and peaks standardized to creatinine (Cr). *Std.* standardization type, *R2X* original data variability, *R2Y* test prediction ability, *Q2* test prediction rate in cross-validation, *NP* unstandardized peaks, *Cr* peaks standardized to creatinine, *pQ2* p-value for Q2Y, *pR2Y* p-value for R2Y.

Std.	Analysis	R2X	R2Y	Q2	No.	pR2Y	pQ2	Sensitivity (95% CI)	Specificity (95% CI)	Accuracy (95% CI)
NA	1.	52.9%	8.5%	0	159	0.05	0.3	55.8% (45.2–65.9)	71.7% (62.1–80)	64.2% (57.1–70.8)
	2.	50.4%	20%	0.09	29	0.05	0.05	67.4% (56.9–76.6)	71.7% (62.1–80)	69.7% (62.8–75.9)
Cr	1.	89.4%	7.7%	0.04	159	0.05	0.05	78.2% (68–86.3)	47.3% (36.9–57.9)	62.2% (54.7–69.3)
	2.	60.2%	20.4%	0.12	28	0.05	0.05	77% (66.8–85.4)	63.4% (52.8–73.2)	70% (62.7–76.6)

Table 2. Results of two OPLS-DA analyzes for unstandardized areas (NA) and areas standardized to creatinine (Cr). (1) analysis - all parameters from the absorption and fluorescence detector; (2) analysis – excluded mutually correlated parameters with $r > 0.9$. *Std.* standardization type, *R2X* original data variability, *R2Y* test prediction ability, *Q2* test prediction rate during cross-validation, *NA* unstandardized areas, *StCr* areas standardized to creatinine, *No.* number of characteristics, *pQ2* p-value for Q2Y, *pR2Y* p-value for R2Y.

Chromatographic analysis

HPLC coupled with absorption and fluorescence detectors allows us to identify metabolites and their changes between groups.

Spectral characteristics (absorption and fluorescence) of urine acquired from control subjects and patients with BC were measured using HPLC on a reversed phase. Out of a total of 179 samples, only 172 samples entered the analysis (missing MCAR data). The HPLC chromatograms were measured for 30 min (Fig. 2).

Analysis of peaks

A total of 210 peaks for the absorption detector (ABS) and 125 peaks for the fluorescence detector were identified. All peaks were also standardized to creatinine. The obtained data entered univariate analysis and ROC analysis.

Univariate analysis Univariate analysis of absorption peaks revealed 37 statistically significant ($p < 0.05$) unstandardized and 27 standardized peaks. Out of 125 fluorescence peaks, 15 (unstandardized) and 8 (standardized) peaks were statistically significant (Supplementary Tables S5–S8).

ROC analysis For both unstandardized and standardized peaks, the accuracy of the ROC analysis for individual peaks ranged from 43 to 75% and from 43 to 72%, respectively. The area under the curve (AUC) values ranged from 0.50 to 0.76 and from 0.50 to 0.73, respectively. The results for the four selected peaks with the highest AUC values are summarized in Supplementary Table S9.

Analysis of areas

As with peaks, areas were standardized to creatinine and subjected to univariate and ROC analysis.

Univariate analysis In the univariate analysis of the absorption areas, 194 out of a total of 480 areas were identified as different between groups ($p < 0.05$). After standardization to creatinine, 93 areas were statistically significant (Supplementary Table S10).

In the univariate analysis of the fluorescence detector, 84 out of a total of 300 unstandardized and 31 standardized areas were different between groups (Supplementary Table S11, $p < 0.05$).

ROC analysis ROC analysis was applied to 60 unstandardized as well as standardized areas of absorption chromatograms for 8 detector settings. Areas with the highest AUC value were selected. For unstandardized areas, the accuracy of ROC analysis was in the range of 49–75%, with sensitivity from 16 to 99%, specificity from 14 to 99% and AUC of 0.5–0.74.

For standardized areas, ROC analysis achieved accuracy of 47–74%, sensitivity of 8–99%, and specificity of 12–100% with AUC of 0.5–0.73.

ROC analysis was also applied to 60 unstandardized as well as standardized areas of fluorescence chromatograms for 5 detector settings. Again, areas with the highest AUC value were selected. For unstandardized areas, the accuracy of the ROC analysis was in the range of 48–73%, sensitivity from 13 to 93%, specificity from 23 to 100% and AUC 0.5–0.76.

For creatinine-standardized areas, ROC analysis achieved accuracy of 47–67%, sensitivity of 8–99%, specificity of 10–100%, and AUC of 0.5–0.66. The results are summarized in Supplementary Table S12.

OPLS-DA Standardized and unstandardized areas were subjected to OPLS-DA analysis in three ways and compared: (1) all absorption and fluorescence areas at all detector settings (780 areas) included; (2) non-colinear areas of more than 0.9 excluded; (3) non-colinear areas of more than 0.8 excluded. Two components (one explanatory and one orthogonal) entered each analysis. The results are presented and compared in Table 3.

Convolutional neural networks Through CNN and data from chromatograms, two models based on urine spectral changes that could distinguish patients with BC from controls were created. In the first case, the model was trained on all samples (training model). In the second case, a 10-fold cross-validation took place (validated

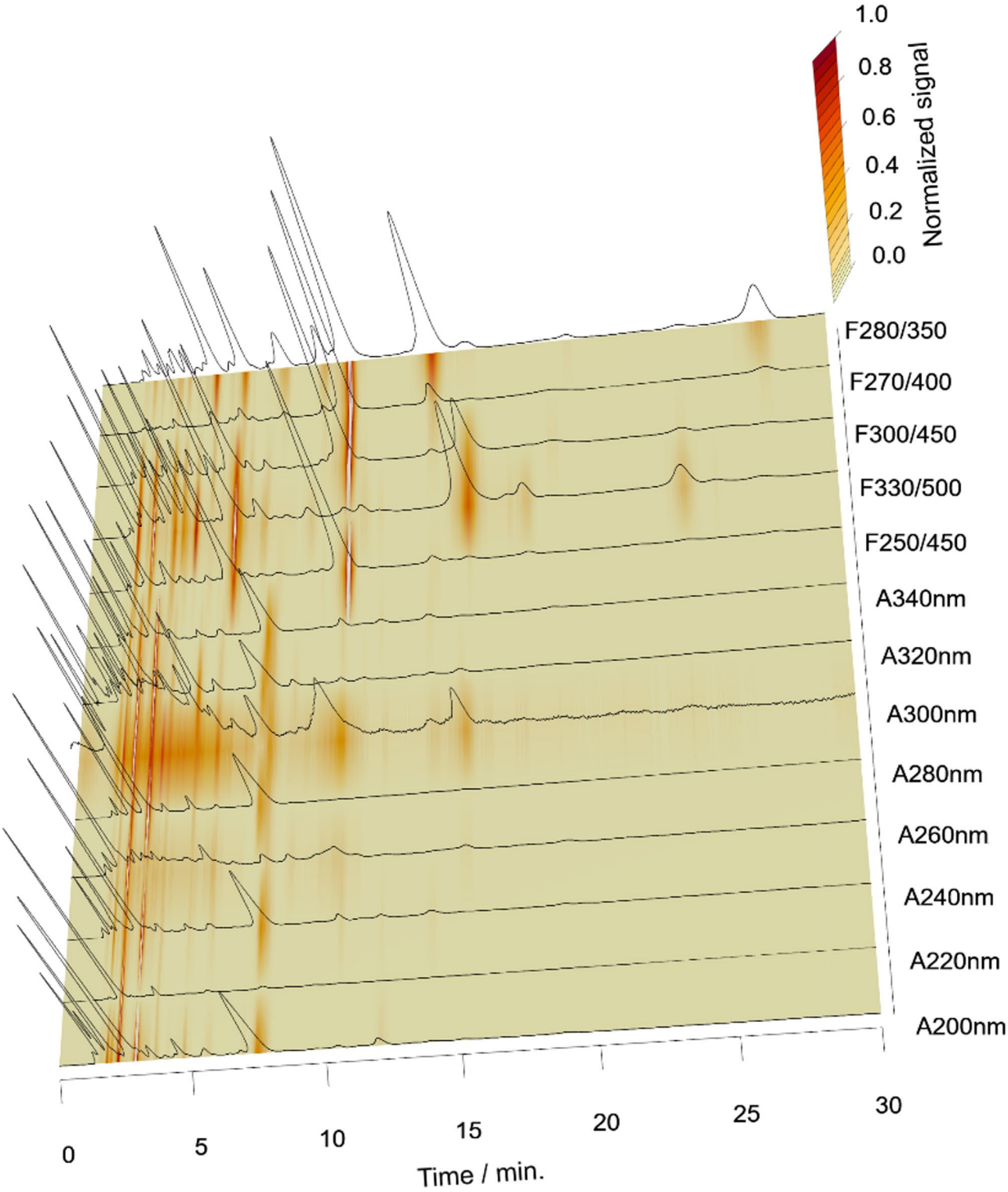

Fig. 2. Representative multiple chromatogram of urine with contour plot. The x-axis represents retention time, the y-axis represents detector settings (A-absorption and wavelength, F-fluorescence and excitation/emission wavelength), and the z-axis represents normalized signal. The contour plot shows the normalized signal using a color scale.

model). The number of epochs in both models was 30. The CNN hyperparameter settings are listed in Supplementary Table S13. For the training model, we achieved an accuracy of 95.3%, with a sensitivity of 97% (95% CI 91–99%) and a specificity of 96% (95% CI 88–99%). With the validated model, the accuracy reached 88.2%, sensitivity 80% (95% CI 71–87%) and specificity 62% (95% CI 48–74%).

Std.	Analysis	R2X	R2Y	Q2	No.	pR2Y	pQ2	Sensitivity (95% CI)	Specificity (95% CI)	Accuracy (95% CI)
NA	1.	19.6%	22.8%	0.02	780	0.3	0.1	54.3% (42.9–65.4)	85.7% (76.8–92.2)	70.9% (63.5–77.6)
	2.	21.5%	30.3%	-0.03	275	0.05	0.15	65.4% (54–75.7)	82.4% (73–89.6)	74.4% (67.2–80.8)
	3.	19.7%	32.7%	-0.02	149	0.05	0.1	67.9% (56.6–77.8)	85.7% (76.8–92.2)	77.3% (70.3–83.3)
Cr	1.	14.3%	29.6%	-0.03	780	0.05	0.2	70.4% (59.2–80)	82.4% (73–89.6)	76.7% (69.7–82.8)
	2.	10.4%	38.2%	0.008	312	0.05	0.05	79% (68.5–87.3)	82.4% (73–89.6)	80.8% (74.1–86.4)
	3.	10.3%	40.9%	0.03	172	0.05	0.05	81.5% (71.3–89.2)	81.3% (71.8–88.7)	81.4% (74.8–86.9)

Table 3. Results of three OPLS-DA analyzes for unstandardized areas (NA) and areas standardized to creatinine (Cr). (1) analysis - all parameters from the absorption and fluorescence detector; (2) analysis - excluded mutually correlated parameters with $r > 0.9$; (3) analysis – excluded mutually correlated parameters with $r > 0.8$. *Std.* standardization type, *R2X* variability of the original data, *R2Y* test prediction ability, *Q2* test prediction rate during cross-validation, *No.* number of characteristics, *pQ2* p-value for Q2Y, *pR2Y* p-value for R2Y.

Discussion

One of the crucial steps leading to successful cancer treatment is early diagnosis, which allows for catching the disease at an early stage[2]. For this purpose, it is important to have sensitive, accurate, and powerful screening and diagnostic examination methods that enable the patient to be examined without fear of a painful and unpleasant procedure. Due to the direct contact, urine provides an image of the state of bladder cells, and any metabolic change in these cells is likely to be reflected in the physicochemical properties of the urine. This could contribute to the development of a non-invasive and useful method for screening and early diagnosis of BC.

Since classical fluorescence spectrophotometry does not allow suitable analysis of multicomponent substances (such as urine), we chose a complex multiple measurement of fluorescence spectra, which allows to create three-dimensional excitation-emission matrices (EEM). The analysis of urine fluorescence has been mentioned in several works as a potential way to reveal peaks or areas that could lead to the creation of a non-invasive diagnostic test[11–13]. We evaluated the obtained data either directly as unstandardized data, or standardized to creatinine, which is commonly used in examinations, especially in cases of a one-time collection[14,15].

Univariate analysis revealed nine peaks, none of which were statistically significant in unstandardized data. Conversely, after standardization to creatinine, all fluorescence peaks were statistically significant, most probably due to higher creatinine levels in bladder cancer group. This fact, and the fact that within one sample all parameters are standardized to the same creatinine value, point together to a correlation of fluorescence characteristics. However, due to this correlation, we cannot unequivocally evaluate the advantage of standardization from these results, as the correlated data provide very little added information. Creatinine correction is the most commonly used method for adjusting urinary dilution; however, it has limitations. Nevertheless, its validity is contingent upon several assumptions that may not universally apply to all metabolites. These assumptions include: (1) consistent production and excretion of creatinine across individuals and over time; (2) absence of biochemical interactions that could confound the relationship between the analyte and creatinine; and (3) a stable mass ratio between creatinine and the analyte across varying degrees of diuresis and urine dilution[16]. Given these limitations, we propose that a combined approach—utilizing both absolute and normalized values—may be more appropriate in biomarker discovery. In the context of machine learning models, normalization is partially managed by the algorithm itself, as urinary creatinine is incorporated as one of the input features.

Univariate analysis revealed which peaks differ between BC patients and controls. However, it does not assess the diagnostic potential of each marker (peak). Therefore, the identified significant peaks were further evaluated using ROC analysis to determine their potential as diagnostic biomarkers. The ROC analysis, which included standardized peaks, reached the highest values in terms of sensitivity, specificity, and accuracy, but with a maximum AUC value of only 0.65, which cannot be considered sufficient. According to[17], an acceptable AUC value should be higher than 0.8. For a biomarker to be considered clinically applicable, the minimum AUC value must reach 0.7[18,19]. The AUC value of 0.65 is therefore insufficient in terms of model performance, however, for verifying the added value of individual and combined markers, this value is, in our opinion, sufficient.

Given the limited discriminatory power of individual markers (peaks/areas) in identifying bladder cancer (BC) patients, we employed advanced machine learning techniques, including logistic regression, OPLS-DA, and convolutional neural networks (CNN). These methods utilize complete datasets (EMM, 3D chromatograms) to construct classification models that go beyond absolute marker values by incorporating inter-marker relationships. This enables the detection of complex metabolic patterns that differentiate BC patients from controls—patterns that are often obscured due to the high dimensionality of the data.

Using logistic regression, we created models with identified fluorescent peaks as the independent variables, which, however, did not achieve sufficient values of sensitivity and specificity. According to multiple sources[20–22], it is known that higher age and male gender increase the probability of BC. For this reason, and due to the high heterogeneity of the studied groups, we adjusted the models for 'age' and 'gender'. However, for unstandardized

data, models were selected based on the lowest BIC value, which did not include any peaks, and discrimination was performed solely based on 'age' and 'gender' variables. From this, we conclude that the fluorescent peaks alone do not have sufficient discriminatory power, but together with the variables 'age' and 'gender', they could represent added value that would increase the performance of the model. On the other hand, too many variables in the regression model can lead to its overfitting[23], which in our case was controlled precisely by the BIC criterion.

As one of the last analyses applied to peaks from urine EEM was OPLS-DA, which is used to identify biologically relevant changes[24]. According to[25], a test prediction rate in cross-validation Q2 > 0 indicates good predictive relevance. High values of Q2 mean good predictive power, while negative values of Q2 indicate that the model has no predictive power or is so-called "overfitted" (overtrained)[26–28]. Although all values of Q2 are higher than 0 (0.07–0.09) in our results, the value is still relatively low. The values of sensitivity, specificity and accuracy were in the range of 62–69% for unstandardized as well as for standardized peaks. Those values are lower compared to the works of[29,30], which, using the same statistical method with Raman spectroscopy, achieved sensitivity and specificity in the range of 77–91%. In general, we can state that not a single method of data standardization increases the predictive ability of the models.

In addition to the peaks, we also evaluated the EEM areas. In the ROC analysis, only areas standardized to creatinine reached AUC > 0.7, which, according to[17,18] could be considered sufficient in terms of clinical applicability and discriminatory power of the model. We performed the OPLS-DA analysis for all variables as well as for non-colinear variables. There are several reasons for the exclusion of highly correlated data - they provide the same information and do not contribute to the predictive ability of the model. The model with too many variables can become overtrained, computational inefficiency might arise, and the difficulty of interpreting increases[31]. The best test prediction rate in cross-validation (Q2 = 0.12) was achieved by the model that included variables from standardized data, from which parameters with $r > 0.9$ were excluded. Compared to OPLS-DA peaks, there was a certain improvement in the predictive power of the model. Our values are already approaching usable values for diagnosis and are comparable to values in other studies[29,30].

Based on previous research[11,13], we hypothesized that obtaining the fluorescent characteristics of body fluids (urine) could be a suitable method for distinguishing patients with cancer. However, our results suggest that distinguishing between the groups of patients with BC and controls was not possible, likely due to the large variability in the number of samples within the groups. The fluorescence of metabolites measured by EEM may have overlapped, as they were measured in a single mixture.

Many studies claim that HPLC is a suitable method for investigating BC[32–37], but they differ when labeling a specific diagnostic biomarker[38]. Therefore, in this analysis, we tried to focus on revealing spectral characteristics that reflect a complex change in metabolism, without specifically naming the biomarker.

Urine chromatograms were previously measured by the HPLC method[12], where patients with bladder cancer and benign hematuria were compared. In the ROC analysis, a peak with the highest value of AUC = 0.82, sensitivity 90% and specificity 74% was identified. In another work, UHPLC-UHRMS was used to differentiate BC patients from controls[39]. Of the 51 identified metabolites, only 5 had an AUC > 0.9 with sensitivity of 81% and specificity of 80%. In our measurements, the best-performing chromatographic peaks reached a maximum AUC of 0.76 and an accuracy of 75%, which indicates insufficient predictive power of individual peaks. We also examined the differences between the areas of the chromatograms, divided into 60 areas (1 area = 30 s of RT). Standardization to creatinine did not enhance the discriminatory power of the model. Models created from areas achieved slightly lower values of sensitivity and specificity than those created from peaks. However, the advantage of using areas is the potential simplification of the automated process for non-invasive BC diagnosis. An increase in predictive power can be achieved by incorporating multiple variables, as demonstrated in study[33], where samples were analyzed using both HILIC and RPLC techniques. By combining two variables, the authors achieved an AUC of 0.96, sensitivity of 93%, and specificity of 97%.

Another multi-target approach, which we also employed, involves the development of discriminative models using OPLS-DA and CNN techniques.

In a study[33], PLS-DA analysis values for selected parameters achieved R2Y = 0.995 and Q2 = 0.75 by combining RPLC and HILIC methods, allowing their model to distinguish two groups of patients with 100% sensitivity and specificity. Džubinská et al. (2021) evaluated OPLS-DA analysis parameters at R2Y = 91.6% and Q2 = 60.8%, with a sensitivity of 100% and specificity of 80%[12]. In an earlier study[32], OPLS-DA was used to discriminate patients with BC, successfully distinguishing all 48 patients using HPLC coupled with hybrid triple-quad time-of-flight mass spectroscopy. Niziol et al. also used OPLS-DA for patient discrimination, showing high values of R2Y = 0.99 and Q2 = 0.96 for the training data set, indicating good interpretability and predictability[39]. OPLS-DA performed similarly on the validation data set (R2Y = 0.984, Q2 = 0.918). Using HPLC-MS, bladder cancer patients ($n = 138$) and controls ($n = 121$, of which 52 controls had benign hematuria) were compared in a study[35], where OPLS-DA model was able to satisfactorily separate these two groups of patients (R2 = 0.88, Q2 = 0.66). In cross-validation, the proposed model had a sensitivity of 91.3% and a specificity of 92.5%, with AUC = 0.94. Compared to these studies, our best OPLS-DA models (sensitivity 81.5%, specificity 81.3%) based on chromatographic peak areas do not have sufficient predictive and discriminatory power and reliability (Q2 = 0.03), regardless of whether the data are standardized or not.

Although the sensitivity and specificity values we obtained in our chromatogram discrimination model can be considered satisfactory (> 80%), in combination with low goodness of fit (R2Y) and goodness of prediction (Q2) values, this model appears insufficient for possible use in non-invasive BC diagnosis. Predictive power of models might be improved by more samples. Since samples are highly variable, this variability may cover the differences between groups related to the presence of cancer. Larger number of samples can eliminate this insignificant variability, providing better information about characteristics related to the presence of malignant disease. The results of the PLS-DA models also indicate that it is necessary to control the number of input

variables to avoid creating overly flexible models, a phenomenon known as overfitting[23]. Based on our Q2 and pQ2 values during validation, it appears that reducing dimensionality increases the reliability of the trained models.

Since convolutional neural networks (CNNs) are usually used in classifications of data obtained by Raman spectroscopy, we decided to apply CNN to evaluate the obtained fluorescence spectra and urine chromatograms. We were inspired by the work of[40], who used CNN to detect changes in the Raman spectrum of tumor tissue after irradiation. The accuracy[34] achieved using CNN was higher compared to common machine learning techniques, including linear discriminant analysis (LDA) and support vector machines (SVM). Our CNN training and validation model based on data from urine EEMs reached high sensitivity (90% and 92%, respectively) and low specificity (51% and 30%, respectively) which indicate reliable detection of positive cases, but it also marked patients without the presence of bladder carcinoma as positive. This is not desirable for screening methods of diagnosis. The model trained on chromatographic data achieved a maximum accuracy of 95.3%, with a sensitivity of 97% and a specificity of 96%. The validation model achieved an accuracy of 88.2%, with a sensitivity of 80% and a specificity of 62%. The maximum values of the CNN model of chromatograms were better than those for EEM, but the sensitivity and specificity in the validation model still did not achieve satisfactory results. Compared to FDA-approved tests, our best-performing CNN and OPLS-DA models based on chromatographic peaks achieved comparable sensitivity and specificity, as FDA tests—depending on the study—often report similar or slightly lower values[4,6]. We could achieve better classification by training a more complex model, but it would be necessary to account for overtraining, which could result in inaccuracies in predictions and problems with predicting new data.

The performance of both CNN and OPLS-DA models has the potential to be improved by increasing the sample size; however, no further improvement is expected from univariate analysis. The limited number of samples represents a key constraint of this study. This is reflected in the high performance observed during training and the comparatively low performance during validation, which is likely due to the small dataset and high inter-patient variability. For such a heterogeneous group (age, gender, stage, grade distribution), a substantially larger number of samples would be required to ensure sufficient information for robust model training. However, acquiring an adequate number of samples is time-consuming and presents a practical challenge.

Conclusion

In the presented work, we aimed to identify differences between patients with BC and controls based on changes in the spectral characteristics of collected urine samples, by several statistical and machine learning methods. First, multiple fluorescent spectra measurements were executed. Various analysis methods were applied to obtained data, but the highest accuracy values were achieved by OPLS-DA and CNN models (≈ 70%). Standardization to creatinine provided the best differentiation of groups, but there was no acceptable increase in the predictive power of any of the models created.

Urine samples were further separated and analyzed using HPLC. Similarly to EEMs, standardization to creatinine did not provide better discriminative models for chromatograms, although it at least reduced the dimensionality of the data, which is desirable for further use in multivariate analyses. By applying CNNs to data obtained from areas of chromatograms, we created a training model with an accuracy of 95.3%; the validated model achieved an accuracy of 88.2%. Combination of chromatographic measurements with CNN could have a potential in future BC diagnostics, however further validation on larger, independent datasets is essential to ensure robustness and generalizability before clinical implementation.

Materials and methods

Urine samples

Samples of urine were provided by the Urology Department of UNB Hospital of St. Cyril and Methodius in Bratislava. A total of 208 urine samples were collected, of which 85 were control samples and 123 were from patients with a preliminary confirmed diagnosis of bladder cancer. The study was approved by the Ethics Committee of the UNB Hospital of St. Cyril and Methodius in Bratislava (125/2021). Samples suitable for analysis came from patients who had signed informed consent forms and met the inclusion and exclusion criteria. The study was conducted in accordance with the Declaration of Helsinki and all study procedures were performed in accordance with the relevant guidelines and regulations. Patients diagnosed with BC (diagnosis C67) according to the International Classification of Diseases ICD 10 were included. Inclusion in the relevant group (patients with cancer or control group) was preceded by histological examination.

Processing of urine samples

Urine samples were stored in a freezer at − 28 °C until processing. Before the analysis they were centrifuged for 20 min at 200 g (1500 RPM, radius 90 mm) and 4 °C. The supernatant was then removed using a NORM-JECT® syringe, and further filtered through an injection microfilter LLG Syringe Filter PTFE with a hydrophobic membrane with a pore size of 0.2 µm.

Characterization of the samples used in fluorescence spectroscopy measurement

A total of 208 samples were measured to acquire multiple fluorescence spectra. From this group, 7 samples were excluded from the analysis due to missing diagnoses. Of the remaining 201 samples, 116 were from patients with confirmed BC, and 85 were controls. There were 125 samples from male patients (mean age 64.2 years) and 76 samples from female patients (mean age 66.1 years). Figure 3A schematically shows the study procedure, and Supplementary Table S14 shows the characterization of the group.

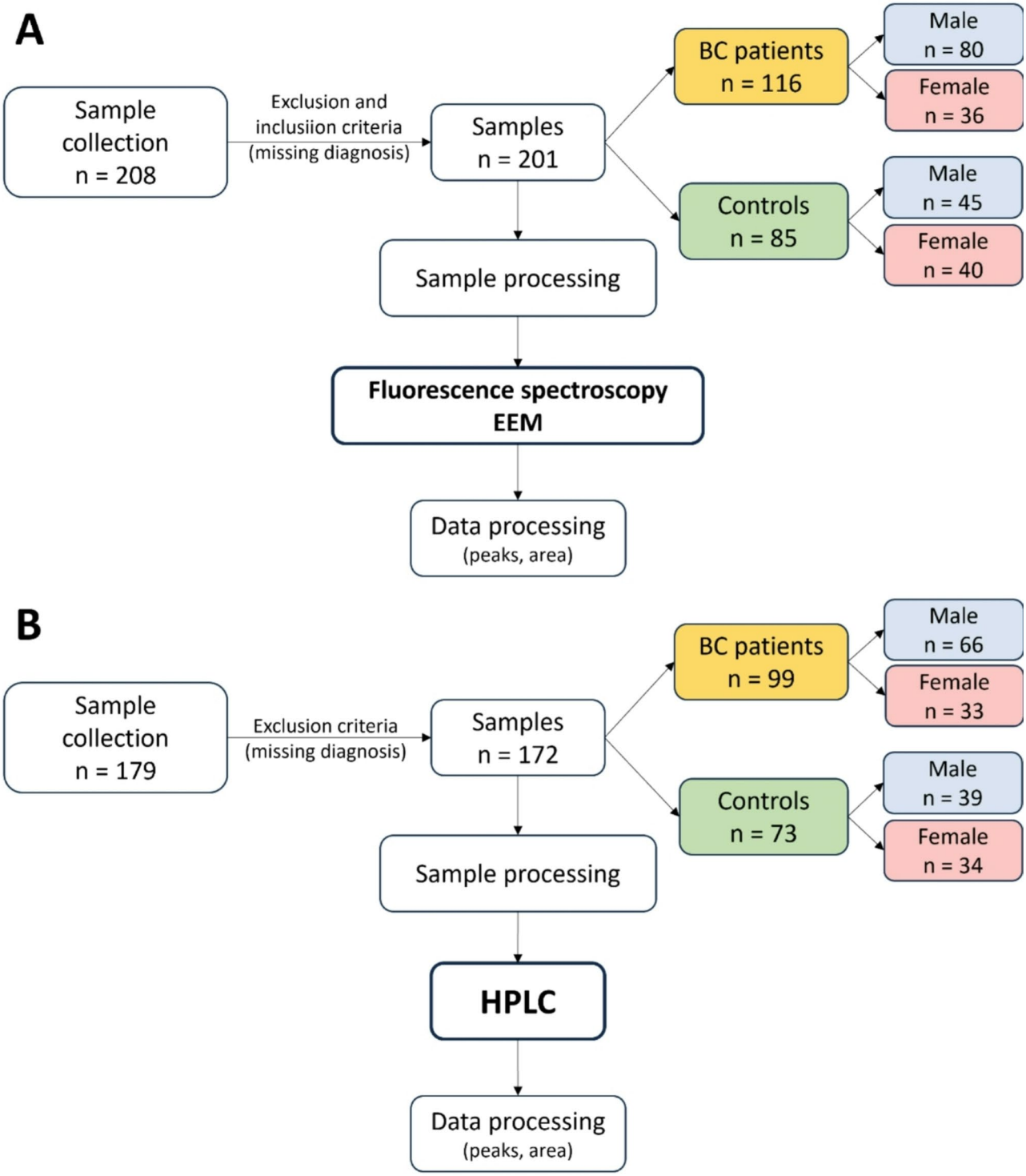

Fig. 3. Schematic illustration of (**A**) the examination process of urine EEM using fluorescence spectroscopy, (**B**) the process of measuring and evaluating urine by HPLC.

Characterization of samples used in HPLC measurements

A total of 179 samples (part of all 208 samples collected) were measured using the HPLC method for absorption and fluorescence chromatograms. Seven samples were excluded due to missing diagnoses. Of the remaining 172 samples, 99 came from patients with confirmed BC, and 73 from controls. There were 105 samples from male subjects (mean age 64.7 years) and 67 samples from female subjects (mean age 66.6 years). Supplementary Table S15 presents the characterization of the group. Figure 3B schematically shows the procedure for measuring and evaluating data obtained from HPLC urine.

Fluorescence spectroscopy

Since common fluorescence spectrophotometry does not allow suitable analysis of multicomponent samples, complex multiple measurements of fluorescence spectra at different wavelengths were executed. Multiple fluorescence spectra were visualized using excitation-emission matrices (EEM). Matrices were measured in excitation and emission wavelengths in the range of 200–800 nm. This range covers both the UV and visible regions.

To determine the fluorescence characteristics of urine, a Perkin Elmer LS 55 fluorescence spectrophotometer was used. The spectra were measured in the wavelength range from 200 to 800 nm at both excitation and emission wavelengths (excitation step 20 nm). Fluorescent peaks from the EEM were selected and compared between the two groups of subjects. Peaks were analyzed using univariate analysis, ROC, logistic regression, and OPLS-DA. EEMs were also divided into fluorescence subareas, which were used for intergroup comparison. The subareas were created by dividing the excitation wavelength axis by 20 nm and the emission wavelength axis by 50 nm, resulting in a matrix divided into 31 × 12 (372) areas with sides of 20 × 50 nm (Supplementary Fig. S2). EEMs were analyzed using univariate analysis, ROC, OPLS-DA, and convolutional neural networks.

Chromatographic analysis

Preparation of the mobile phase

To measure the spectral characteristics of urine, a phosphate buffer solution of potassium dihydrogen phosphate (KH_2PO_4) was used, with a concentration of 10 mM and 5% methanol, pH 6.8. The mobile phase was filtered through a filter with a pore size of 0.45 µm. For spectral analysis of the samples, we used the Prominence 20 A HPLC system, which enables absorption and fluorescence detection of the separated components. A Nucleosil 100-5 column with a C18 stationary phase was used at a constant temperature of 25 °C.

Measurement parameters of HPLC

During the entire measurement, a constant flow of the mobile phase through the column was set at 1 ml/min. The volume of used samples was chosen based on the obtained signal. The absorption detector was set to λ = 200–340 nm with a step of 20 nm to cover the absorption spectrum. Each sample was measured five times at different absorbance and fluorescence detector settings, labeling each setting as "method" 1–5. Supplementary Table S16 presents the settings of the absorption and fluorescence detectors together with the volume of the injected sample.

Chromatograms were measured for a total of 30 min. When analyzing the chromatograms using the LC Solution software, appropriate absorption and fluorescence chromatograms with the highest number of peaks (and thus the best separation of individual components) were selected. Statistical analysis was performed using univariate analysis and ROC analysis. Similar to fluorescence spectroscopy, this period was divided into 60 areas (each lasting 0.5 min). Areas were also subjected to univariate and ROC analysis. Furthermore, machine learning tools, such as OPLS-DA and convolutional neural networks, were used.

Statistical analysis

To identify significant peaks and areas from both fluorescence spectroscopy and HPLC measurements, univariate analysis using the Student t test, or in case of violation of the assumptions of normal distribution, the non-parametric alternative Mann-Whitney U test was performed. Significant peaks and areas were also subjected to ROC analysis.

The dividing point in the ROC analysis was selected using the "Youden" method[41]. After ROC analysis, sensitivity, specificity, accuracy, and AUC were further calculated. Independent variables were evaluated in the same way after standardization for creatinine and total fluorescence.

Multivariate statistical analysis (machine learning)

Due to the complexity of data obtained from spectroscopic measurements, various methods of multivariate analysis were applied to the data. Logistic regression, OPLS-DA and CNN were applied to independent variables (peaks and areas from EEM and chromatograms). Figure 4 shows which statistical analysis was used on which independent variable.

Logistic regression was chosen as the primary modeling technique due to its suitability for binary outcome prediction and its interpretability in terms of odds ratios. To determine which predictor variables to include in the model, an exhaustive approach was applied, systematically evaluating all possible combinations of input variables. This method ensures a comprehensive and unbiased assessment of potential models. The Bayesian Information Criterion (BIC) was used to select the optimal model. BIC is particularly effective for model selection as it penalizes the inclusion of additional parameters, thereby discouraging overly complex models and reducing the risk of overfitting. As a result, the final model reflects the best balance between explanatory power and simplicity[42].

OPLS-DA (Orthogonal Partial Least Squares Discriminant Analysis) is a supervised multivariate method used to identify variables that best separate predefined groups, such as patients and controls. It improves interpretability by separating predictive variation (related to group differences) from orthogonal variation (unrelated noise). To ensure the reliability of the model, cross-validation is used to estimate its performance on unseen data, while permutation testing assesses whether the observed separation is statistically significant or could have occurred by chance. A Q^2 value above 0.5 is generally considered acceptable for good model predictability, while Q^2 below 0.5 raises concerns about overfitting and reduced reliability[43].

Convolutional neural networks (CNN) were applied to data from EEM areas and urine chromatograms. Architecture design and parameter optimization were inspired by the work of[34]. The model was validated using 10-fold cross-validation, which involves creating 10 models, where for each model the data from EEM (or

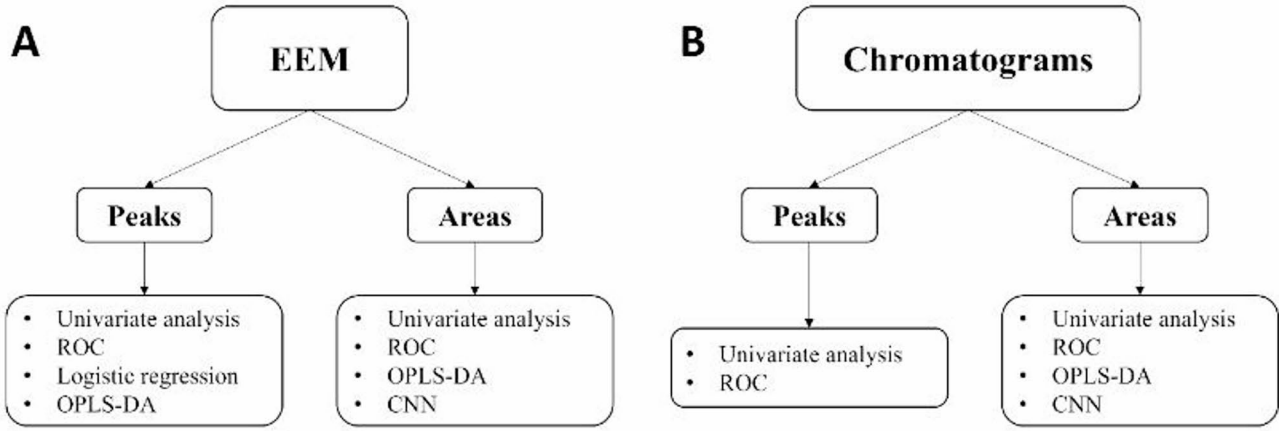

Fig. 4. Applied statistical methods for peaks and areas identified in (**A**) EEM and (**B**) chromatograms.

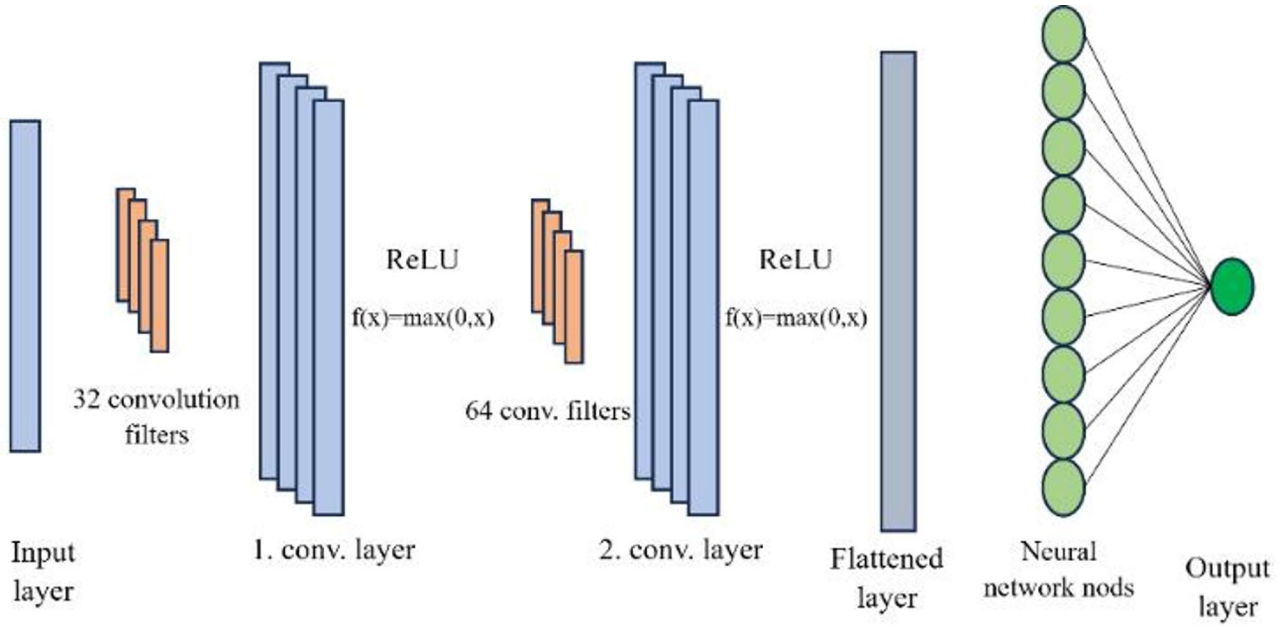

Fig. 5. CNN architecture for the analysis of EEM and urine chromatograms.

chromatograms) were randomly divided into training and validation groups in a ratio of 9:1[23]. We tried different combinations of number of layers, sizes and number of convolutional filters, activation functions, and learning rate and selected the most suitable combination that resulted in the model with the highest sensitivity, specificity, and accuracy. A schematic representation of the CNN architecture is shown in Fig. 5.

The illustration begins with an input layer that receives raw image data (e.g. EMM or chromatogram). This is followed by a first convolutional layer consisting of 32 filters, which extract basic features such as edges and textures from the input. A ReLU (Rectified Linear Unit) activation function is then applied to introduce non-linearity by zeroing out negative values. The output is passed to a second convolutional layer with 64 filters, which captures more complex and abstract patterns. Another ReLU activation follows to maintain non-linearity. The resulting multi-dimensional feature maps are then flattened into a one-dimensional vector, preparing the data for the fully connected layers. These dense layers combine the extracted features to make predictions, and the final output layer produces the network's prediction, typically in the form of class probabilities for classification tasks. The values of the convolutional filters are not manually set, but they are learned automatically by the CNN during the training process through backpropagation, allowing the network to optimize feature detection based on the data.

Software

To evaluate the data from the chromatograms, we used LC Solution version 1.25, which made it possible to compare the measured chromatograms. EEMs were recorded using the BL Studio program.

For further data processing and statistical analysis, we used the R software with used the "pROC" library for ROC analysis[44], and the "glmulti" library for logistic regression[42]. The "ropls" library[45] was used for the (O)PLS-DA method, and the "tensorflow"[46] and "keras"[47] libraries were used for the neural networks.

Data availability

Data that support the findings of this study are available on the following link: https://figshare.com/ndownloader/files/48841420.

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Author contributions
M.V.: Data acquisition and curation, Investigation, Funding acquisition, Writing – original draft. D.M.: Data acquisition. I.W.: Funding acquisition, Supervision. B.K.: Project administration, Supervision. J.J.: Data acquisition and curation. Ľ.H.: Funding acquisition, Supervision. M.Z.: Conceptualization, Formal analysis, Methodology, Writing – review & editing.

Declarations

Competing interests
The authors declare no competing interests.

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